



Introduction to Computer Vision & Real-World AI Challenges

Course Structure

- S1 - Introduction to Computer Vision & Real-World AI Challenges
- S2 - Image Representation & Digital Image Fundamentals
- S3 - Data Preparation for Machine Learning Projects
- S4 - Filters & Convolution (Foundations of Deep Learning for Vision)

MID TERM EXAM (19/10)

- S5 - Classical Machine Learning for Vision
- S6 - Artificial Neural Networks (ANNs)
- S7 - Convolutional Neural Networks (CNNs)
- S8 - Vision Tasks: Classification, Detection & Segmentation
- S9 - Vision Transformers & Modern Architectures
- S10 - End-to-End AI System, Deployment & Multimodal Interface (Image + Speech)

FINAL EXAM (30/11)

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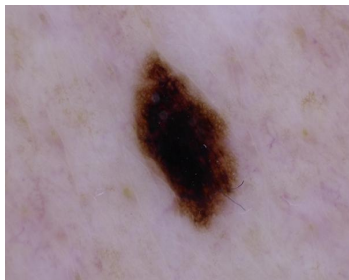
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Learning Objectives

- Define **computer vision** and explain its **goal**
- Explain how an **image** is **formed** and **represented digitally**, and relate this process to the **biological visual system**.
- Recognize the **main computer vision tasks** — classification, object detection, and segmentation.
- Explore and characterize a **real-world image dataset (MILK10k)**.
- Load, inspect, and **visualize MILK10k** programmatically in **Python**.

PROJECT



Real MILK10k dermoscopic image



Malignant or benign?

Today's Agenda

PART 1 — THEORY

80 min

BREAK

20 min

PART 2 — HANDS-ON PYTHON

80 min

o Part 1 (Theory)

- o Vision
- o Image formation
- o Human vision
- o From biological vision to computer vision
- o Computer Vision
- o Measuring light vs. Scene Properties
- o Image: a formal definition
- o Computer Vision Tasks
- o ML Challenge
- o MILK10k dataset

o Part 2 (Practice)

- o Set up Python
- o Load MILK10k metadata and images
- o Explore the dataset and class balance
- o Build reusable image-grid viewer
- o Homework:
 - o Choose a code editor (e.g., VS Code)
 - o Create a GitHub repository for the course homework and final project



SESSION 1

Part 1 — Theory

Image

Let's start with something very simple: an image.



Total solar eclipse, August 12, 2026

Image

Let's start with something very simple: an image.

It's electromagnetic radiation that travelled roughly 150 million kilometres, entered a camera, interacted with a sensor, and was converted into numbers. And somehow, from those numbers, we see this eclipse.



Total solar eclipse, August 12, 2026

Vision

? *Every image starts as energy reaching a sensor – what kinds of energy can form one?*



Total solar eclipse, August 12, 2026

Vision

- ? *Every image starts as energy reaching a sensor – what kinds of energy can form one?*
- o Sunlight is one of the fundamental sources of information about our world.
- o Almost every living organism has some way to sense light. How do we humans do it?



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- **Light interacts with the world → reflects toward our eyes → reaches the retina → our brain interprets it as a visual scene.**



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The light arriving at our eyes doesn't come with labels saying 'this is an eclipse'. It is simply electromagnetic radiation carrying information about the scene. Our visual system transforms those signals into an interpretation of the world.

Vision

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And that is essentially the problem that **computer vision** addresses:
how can we make a machine extract meaningful information from visual data?

Where Do Images Come From?

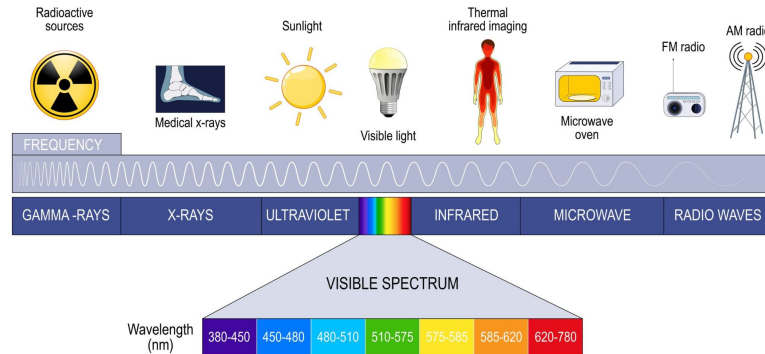
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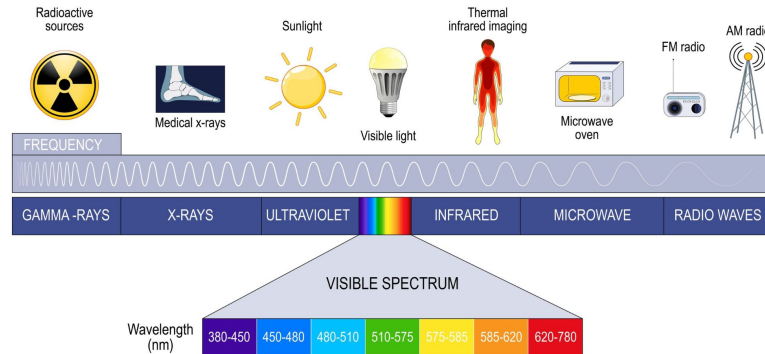
Electromagnetic spectrum



Where Do Images Come From?

- An image is a spatial measurement of some physical phenomenon.
- The most familiar way to form an image is by detecting **electromagnetic radiation**.
 - **Visible light**: photography, human vision. It is just one small part of the electromagnetic spectrum.
 - **Gamma rays**: nuclear medicine, astronomy.
 - **X-rays**: medical imaging, industrial inspection, astronomy.
 - **Ultraviolet (UV)**: fluorescence microscopy, astronomy, remote sensing.
 - **Infrared (IR)**: thermal imaging, remote sensing, astronomy, night vision.
 - **Radio waves**: radio astronomy, radar.

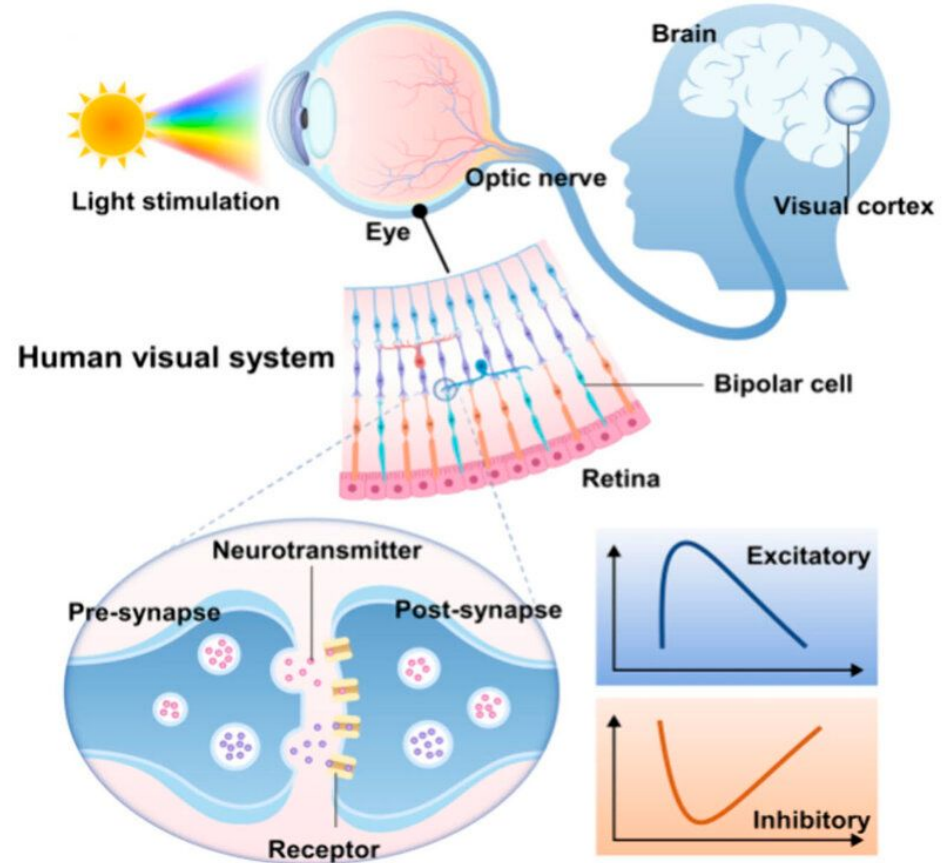
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Some other physical phenomenon sources:

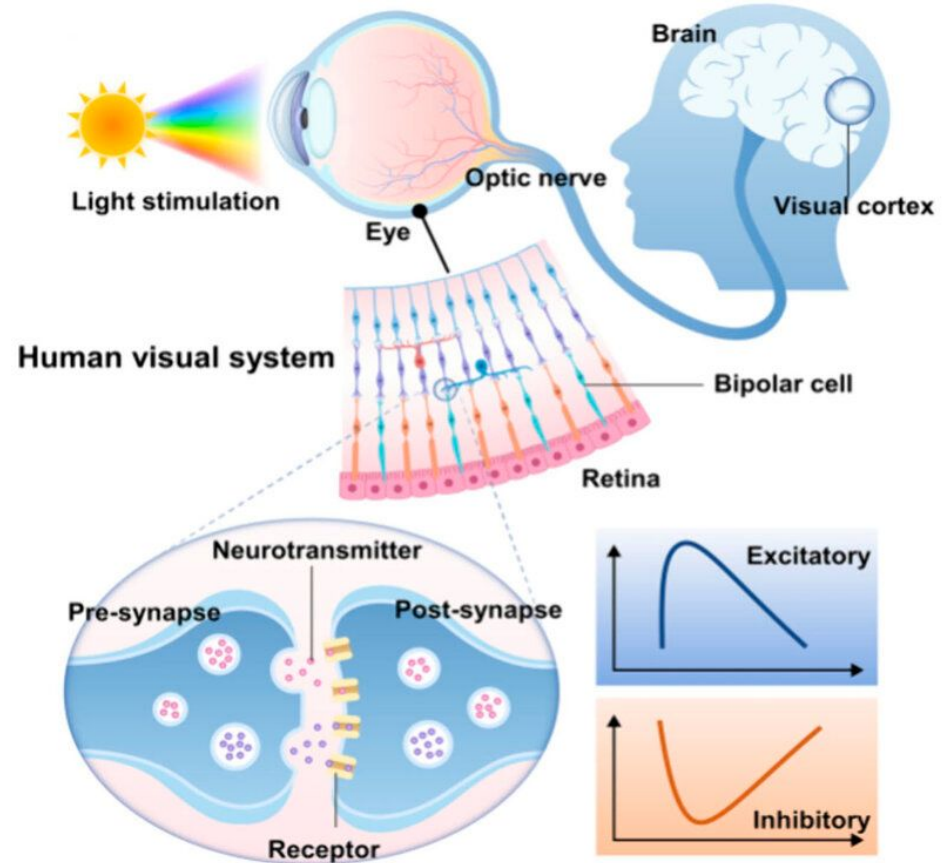
- **Sound / acoustic waves**: ultrasound imaging
- **Electron interactions**: electron microscopy
- **Seismic waves**: seismic imaging of Earth's interior, geophysical exploration

The Human Vision Pipeline



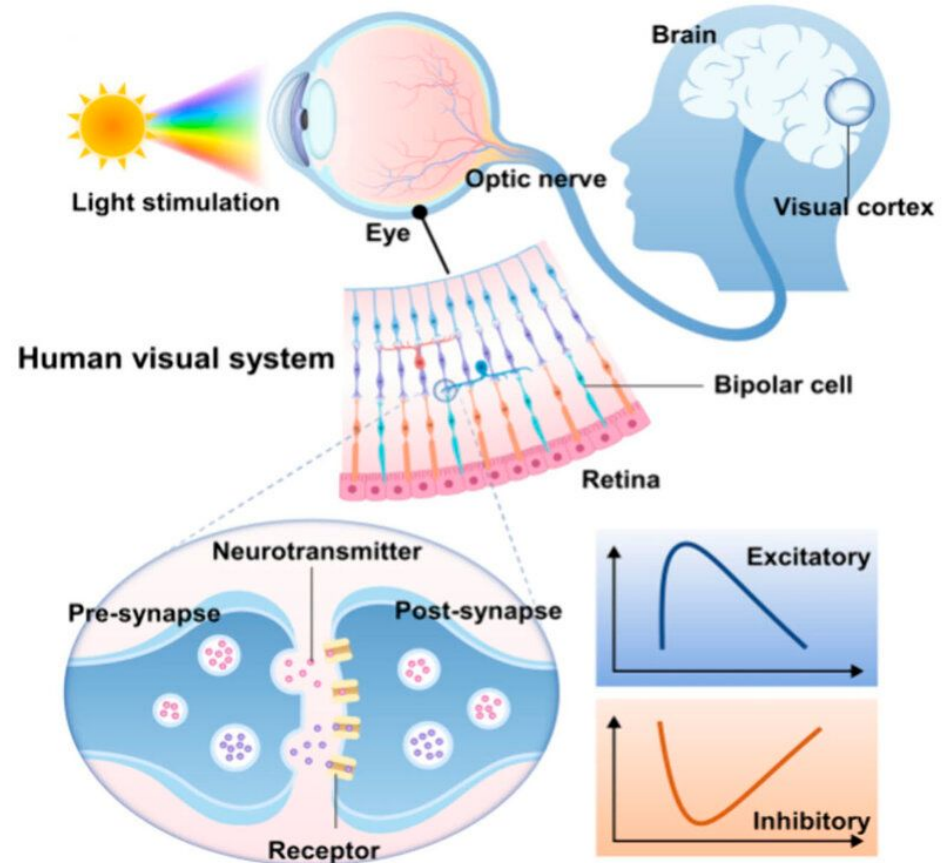
The Human Vision Pipeline

- Light enters the eye and is focused by the lens onto the **retina**, a light-sensitive layer at the back of the eye.



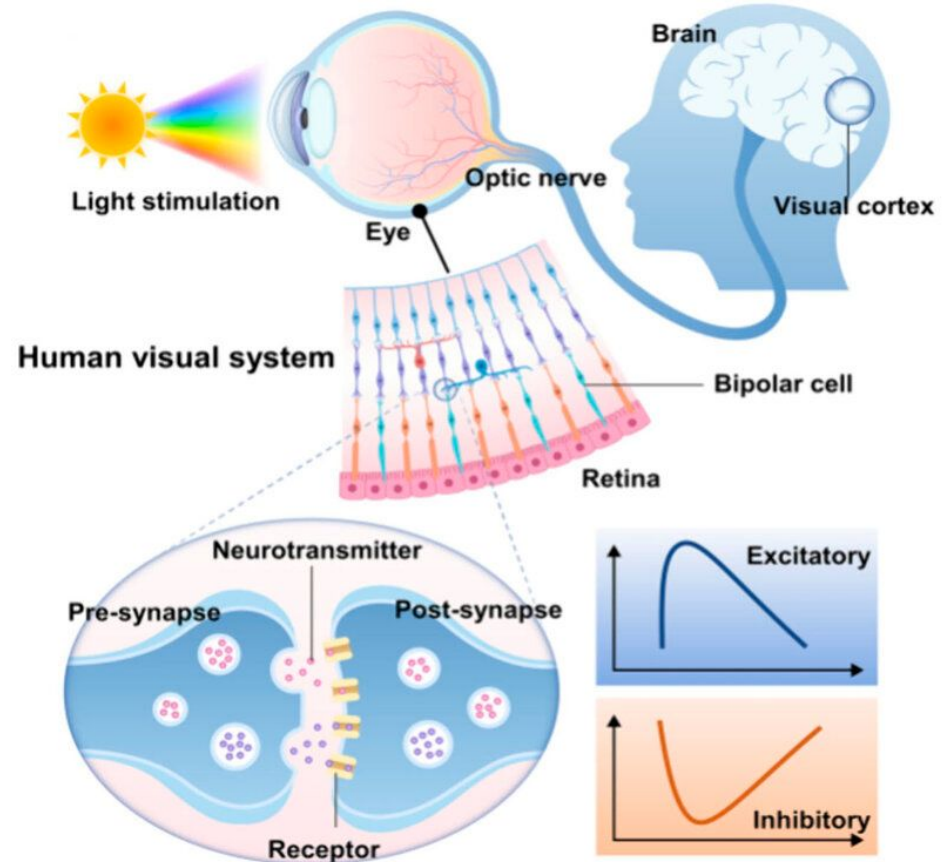
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- The retina contains around 120 million **photoreceptors**: **rods** (sensitive to low light, no color) and **cones** (three types, sensitive to color).



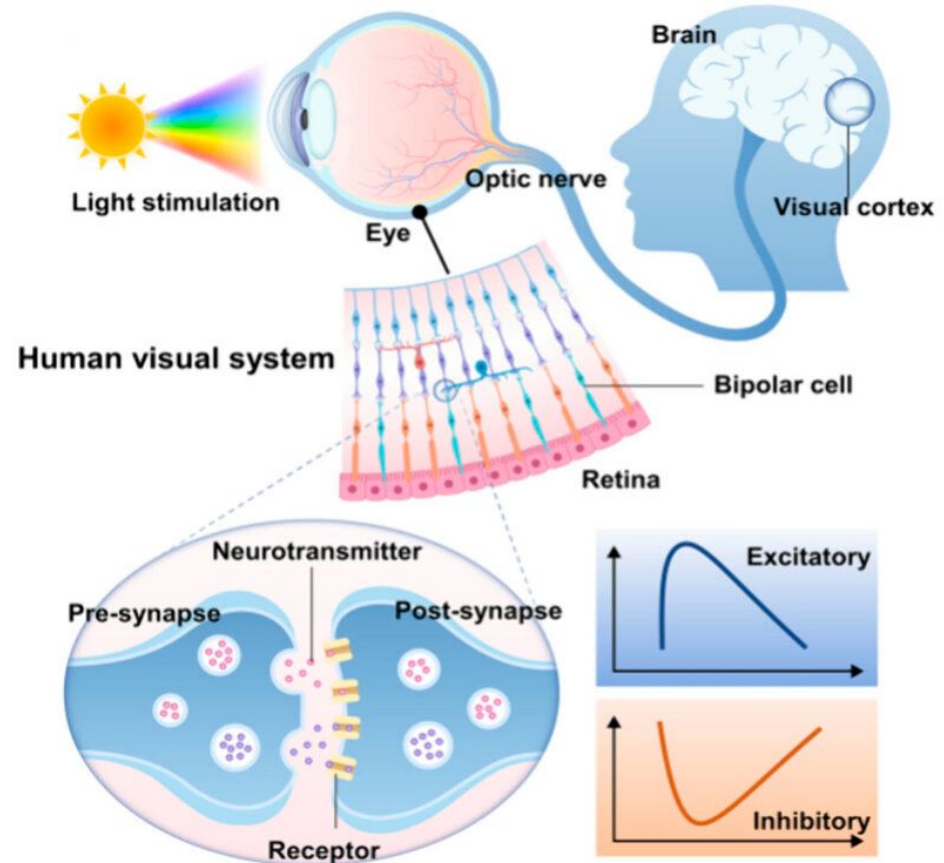
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- Photoreceptors convert **light into neural signals**. These signals are processed through several stages of the visual system and eventually reach the visual cortex, where information about features such as contrast, edges, shapes, motion, and objects is extracted and combined.



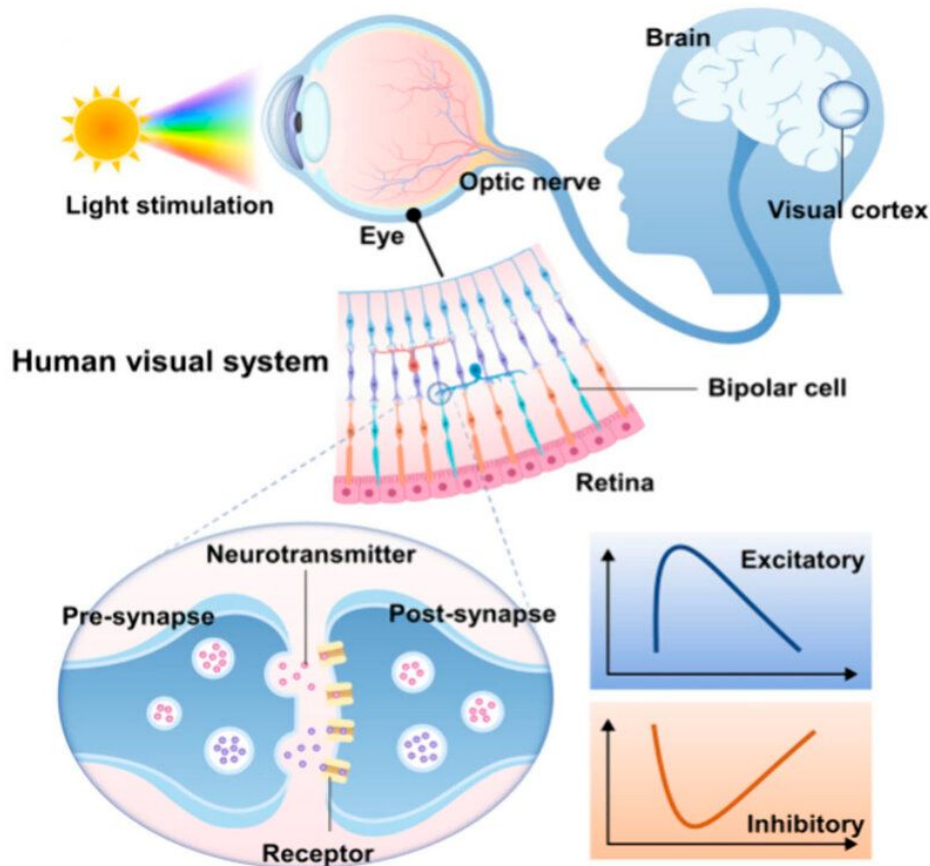
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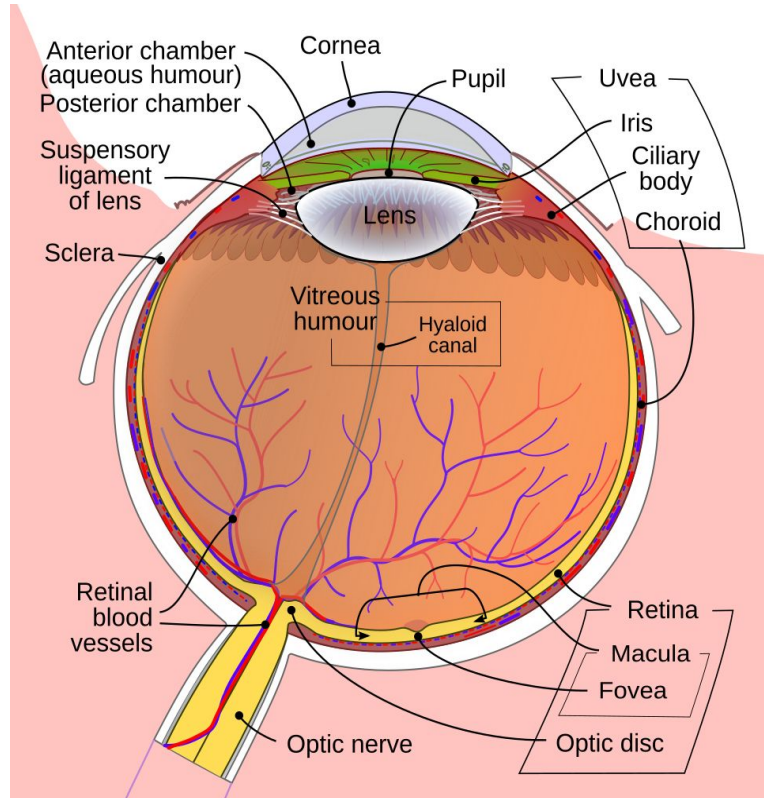
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- **Visual processing is hierarchical**: representations at different stages become increasingly complex and selective.
- This idea of hierarchical visual representations inspired **Convolutional Neural Networks (CNNs)**.



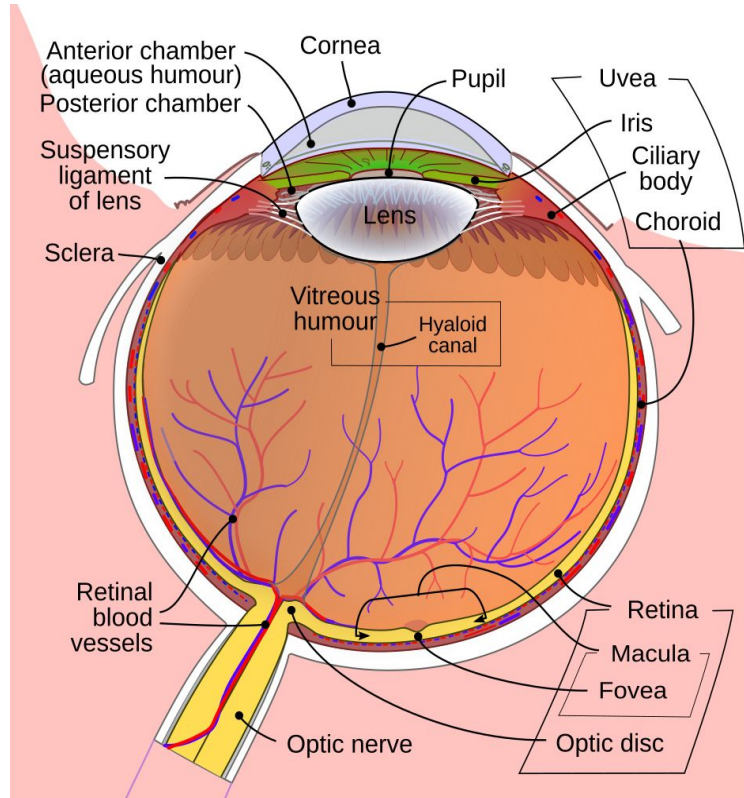
The Eye: Our Biological Camera

- The **eye's optical system** — primarily the cornea and lens, with the iris controlling how much light enters — **focuses** incoming **light onto the retina**.
- The **retina converts light into neural signals** and performs some initial processing before information is transmitted to the brain.
- The **fovea** is a small region near the center of the retina with a very high density of cones. It provides our highest spatial acuity and finest visual detail.
- Outside the fovea, photoreceptor density decreases, so our visual acuity is lower. Human vision is therefore not equally detailed across the visual field.
- The **optic disc**, where the optic nerve leaves the eye, **contains no photoreceptors**. This **creates a blind spot**.



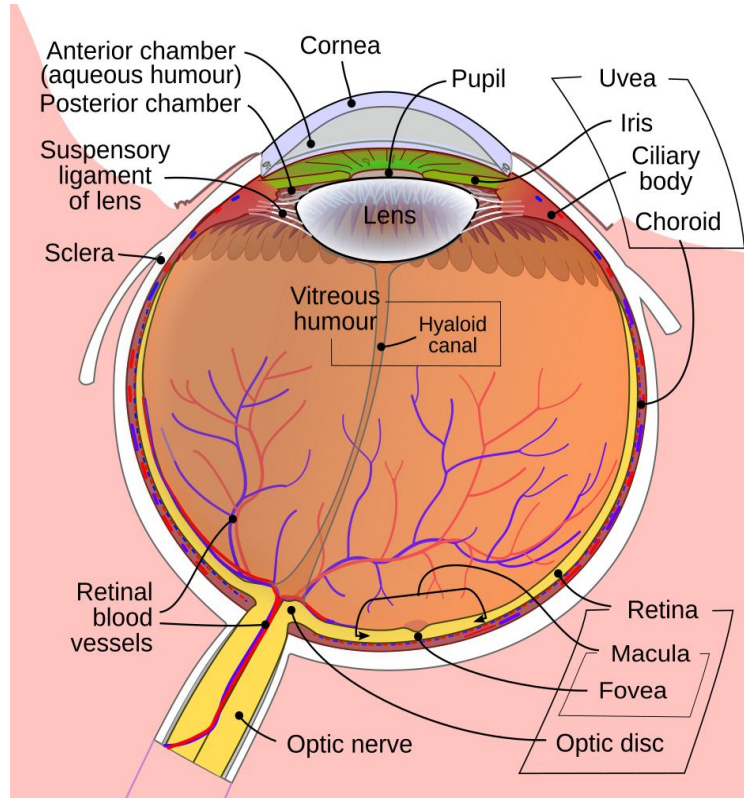
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The Eye: Our Biological Camera

- If there are no photoreceptors here, why don't we normally see a black hole in our vision?
- Because the **brain fills** in the **missing information** using the surrounding visual context.
- So even human vision isn't simply recording everything that reaches the eye. **Information** is **sampled unevenly, processed,** and **sometimes even inferred.**



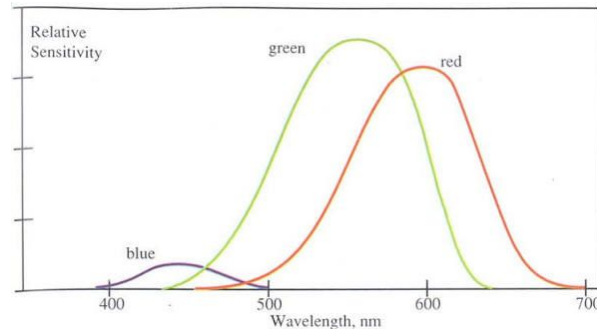
The Retina: Rods and Cones

The retina contains two main types of photoreceptors with different roles:

Type	Count	Color sensitive?	Sensitivity	Response speed
Cones	~5 million	Yes (3 types)	Low (less numerous)	Fast
Rods	~100 million	No	High (more numerous)	Slow

The three cone types respond differently across wavelength — loosely 'red', 'green', and 'blue' sensitive — with the blue-sensitive cones the rarest, at under 10% of all cones. Their sensitivity curves overlap substantially.

The brain compares the responses of these three cone types to infer color.

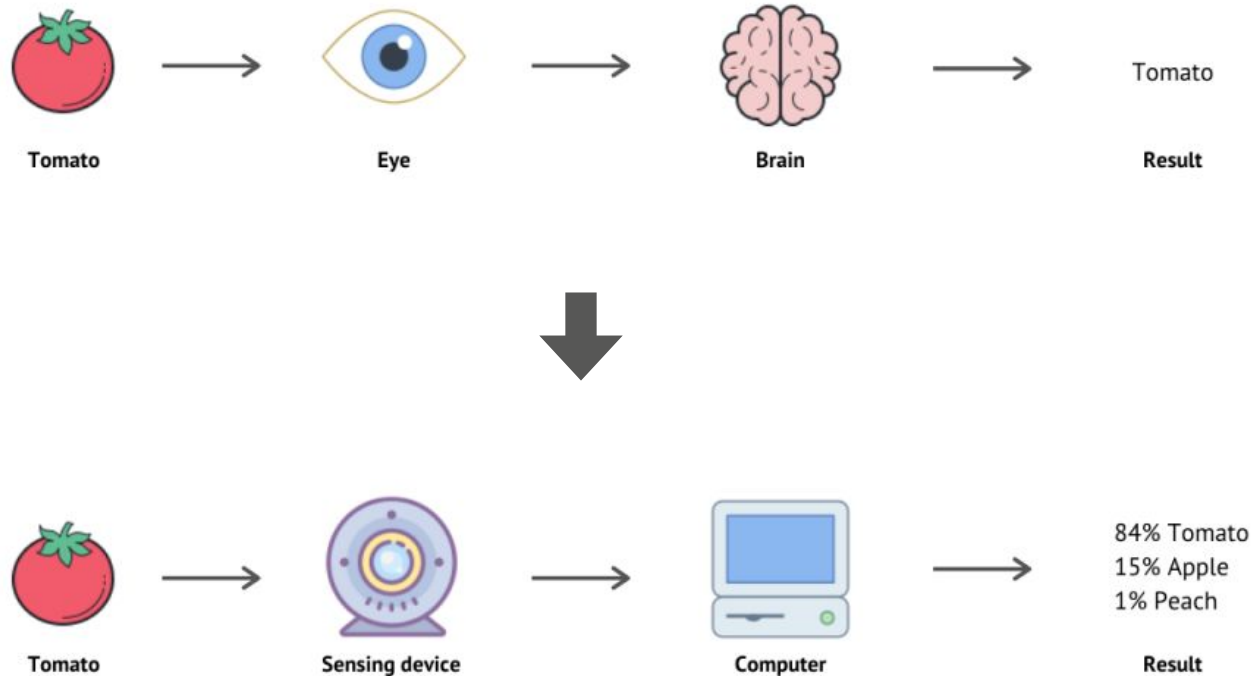


From Biological Vision to Computer Vision

- We have a biological system that turns light into an internal representation of the world; can we build an **artificial system** that **does something analogous**?

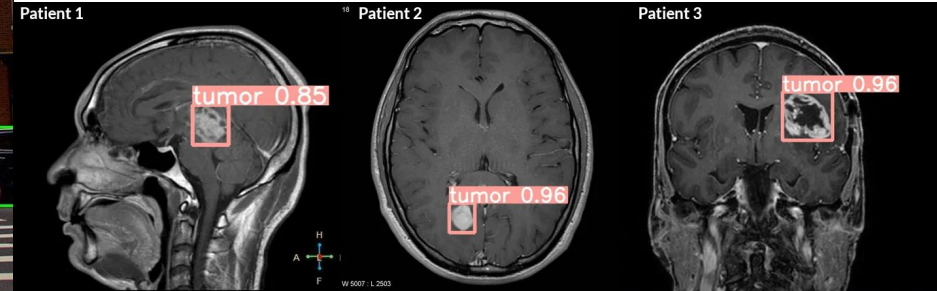
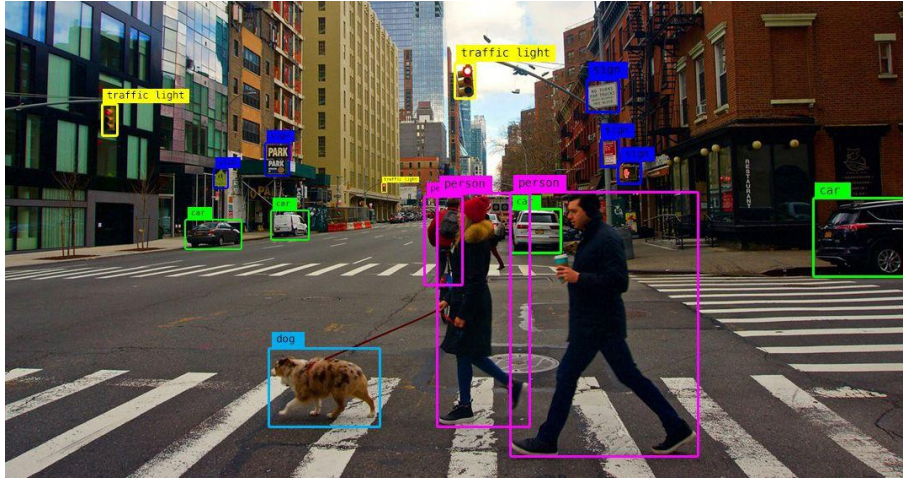
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What Is Computer Vision?

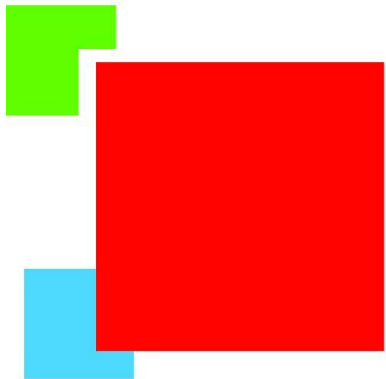
- Computer Vision (CV) is the field of AI that enables machines to extract information and meaning from visual data.
- Since its origins, CV has been an interdisciplinary field spanning computer science, mathematics, physics, biology, neuroscience, psychology, and engineering.



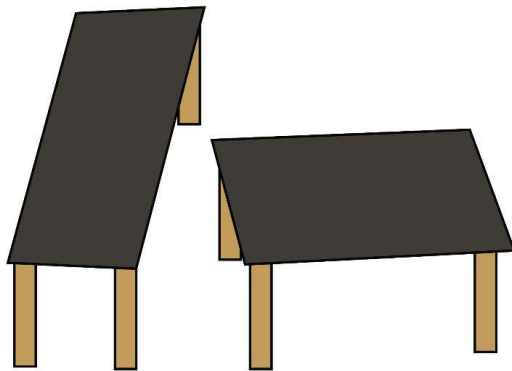
Object detection in a street scene and tumor detection in MRI scans.

Measuring Light vs. Measuring Scene Properties

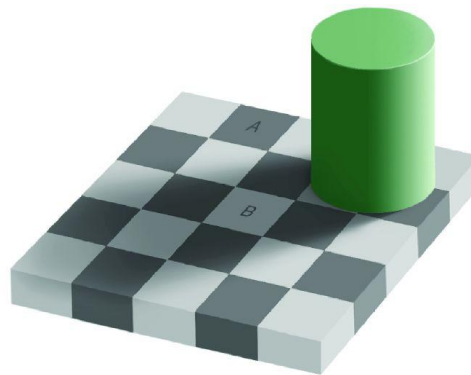
- What do we mean by extracting meaning? For instance, what do we see below?



(a)



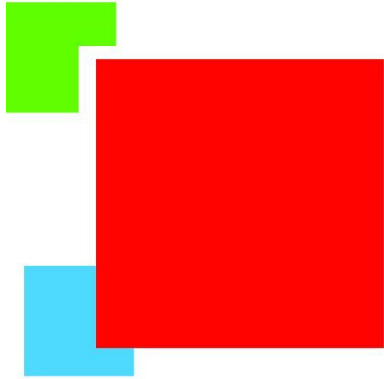
(b)



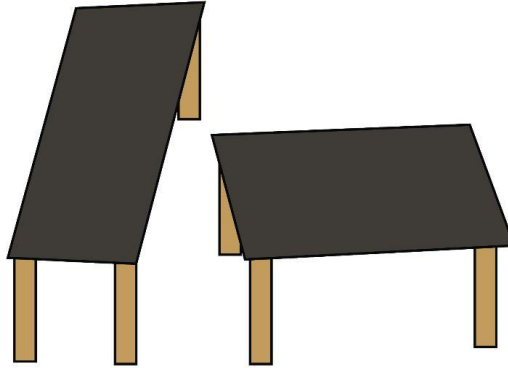
(c)

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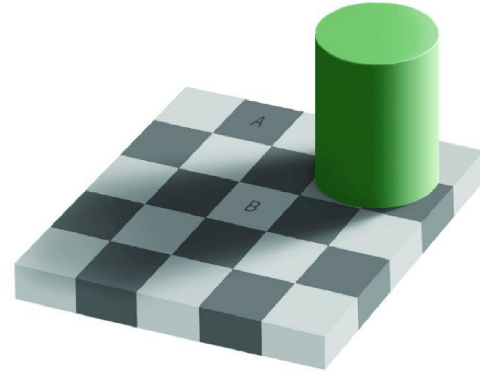
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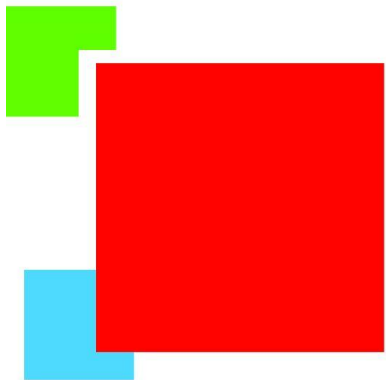
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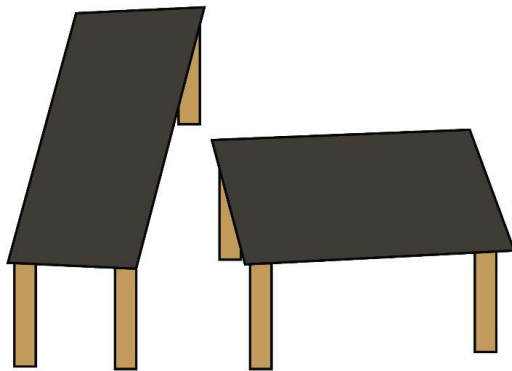
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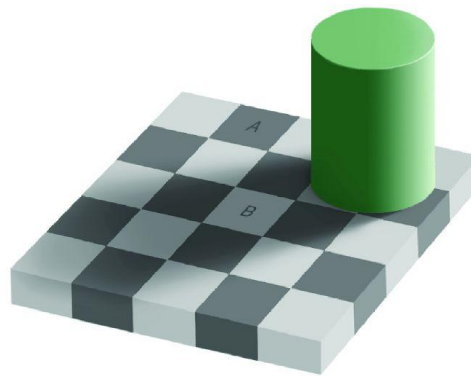
- What do we mean by extracting meaning? For instance, what do we see below? We perceive objects, surfaces, depth, shadows, materials, etc.
- Vision is an inference problem: What properties of the world could have produced these measurements?



(a)



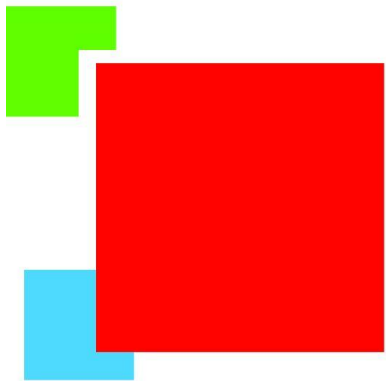
(b)



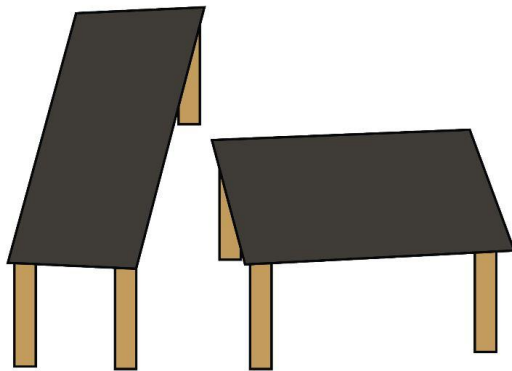
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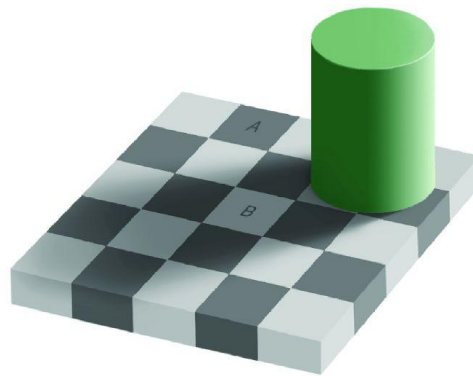
- **(a) Occlusion:** we perceive a red square occluding a smaller blue square — not an L-shaped figure — because squares and occlusions are common in the world.



(a)



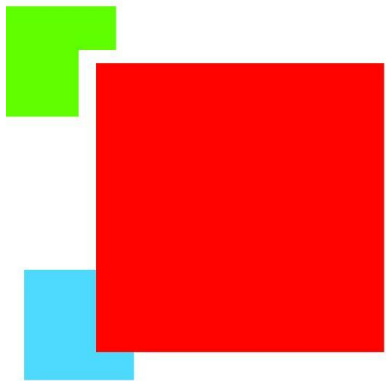
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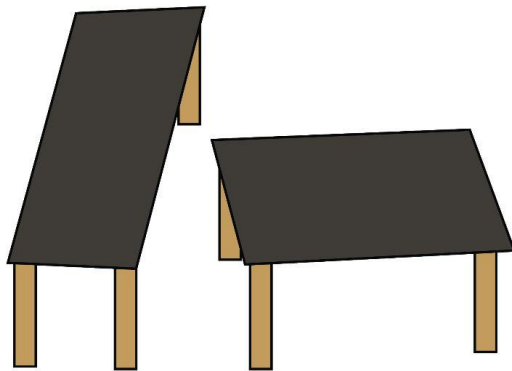
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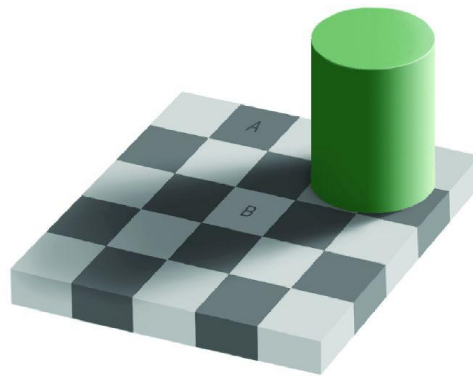
- **(b) Turning-tables illusion:** both tabletops are identical in shape and size, yet the visual system insists on a 3D interpretation, making the left table look longer — even once we know how the image was generated.



(a)



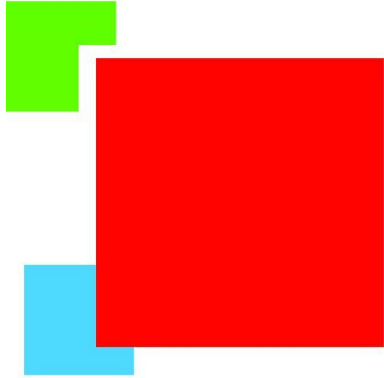
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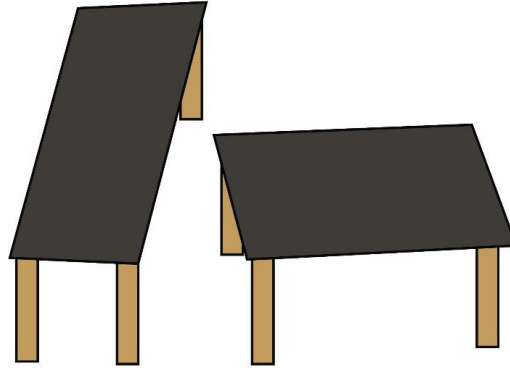
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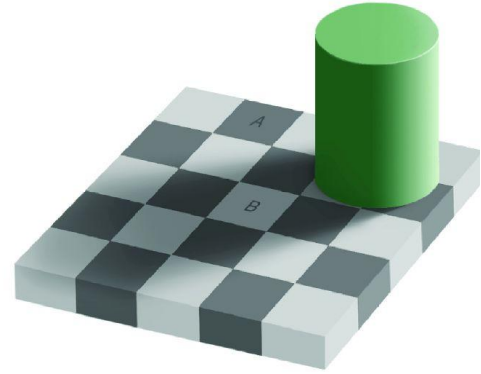
- (c) **Checkershadow illusion:** squares A and B share the exact same pixel intensity are perceived as different shades, because the visual system compensates for a shadow crossing the scene.



(a)



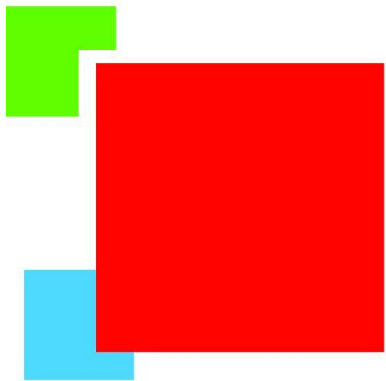
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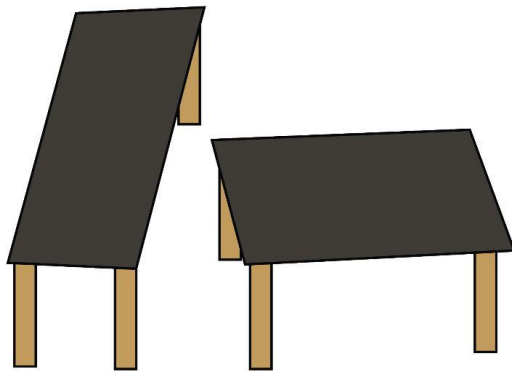
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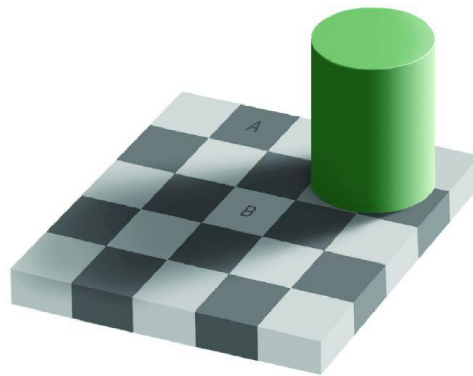
- The goal of vision is **not** to measure light intensities, but to extract **scene properties** relevant to the observer.
- Classic visual illusions show how the human visual system tries to recover the **scene that caused the image**, not just record raw light.



(a)



(b)



(c)

(a) Occlusion — (b) Turning-tables illusion — (c) Checkershadow illusion

Takeaway

Perception $\neq$ raw pixels. Vision systems — human or artificial — must infer meaning, not just measure light.

What Can a Computer Vision System Do?

- What Can Computer Vision Do?

What Can a Computer Vision System Do?

- **Understand what is there**
 - object recognition
 - scene recognition
 - material recognition
 - reading text
- **Understand where things are**
 - object localization
 - segmentation
- **Understand how things change**
 - motion estimation
 - tracking
- **Reconstruct the world**
 - depth estimation
 - 3D reconstruction

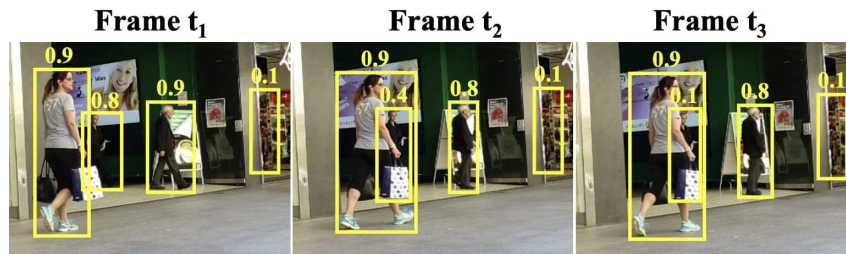
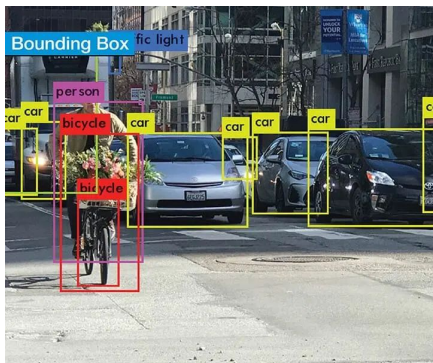


Image: A Formal Definition

But before a computer can solve any of these problems, we need to understand what the computer actually receives.

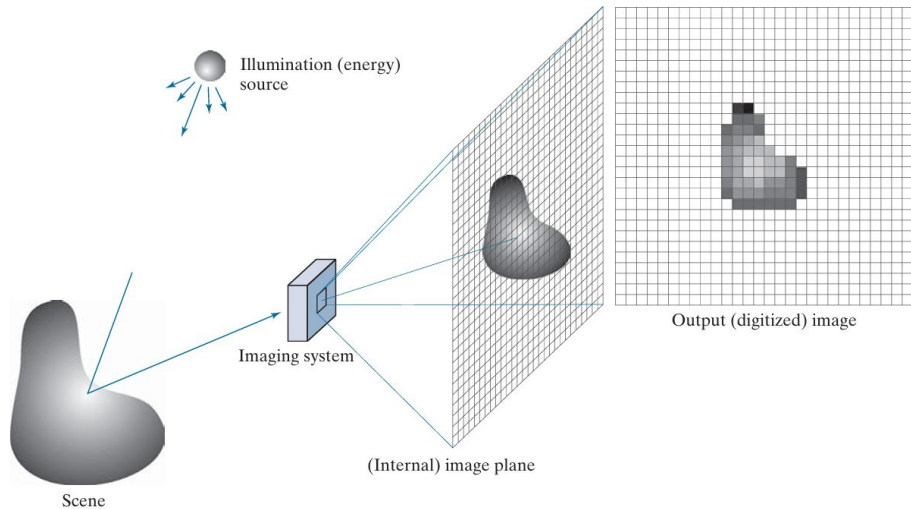


Image formation with a digital camera: light reflects off a scene, a lens focuses it, the sensor converts photons into electrical charge, an ADC digitizes that signal, and the result is a pixel array — the digital image.

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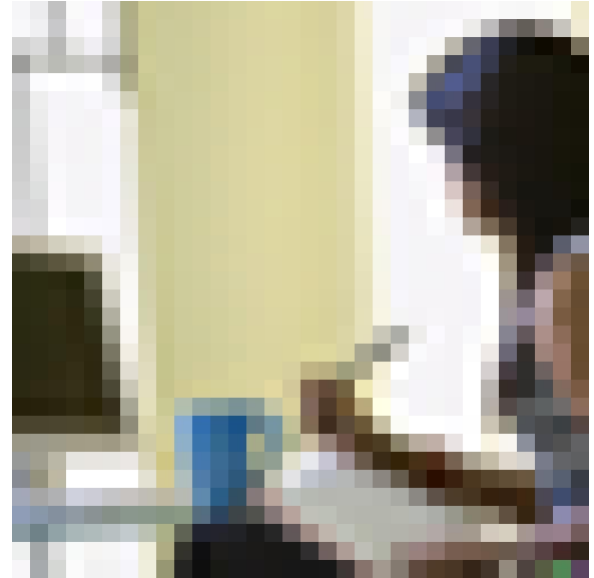
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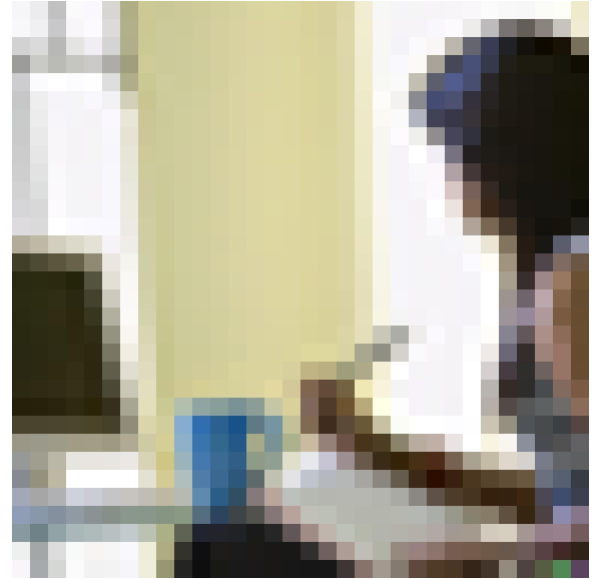


A 32×32-pixel image

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- A **digital image** samples this function at a **finite number of locations** and represents the measurements with **finite numerical values**: a 2D array arranged in rows and columns.



A 32×32-pixel image

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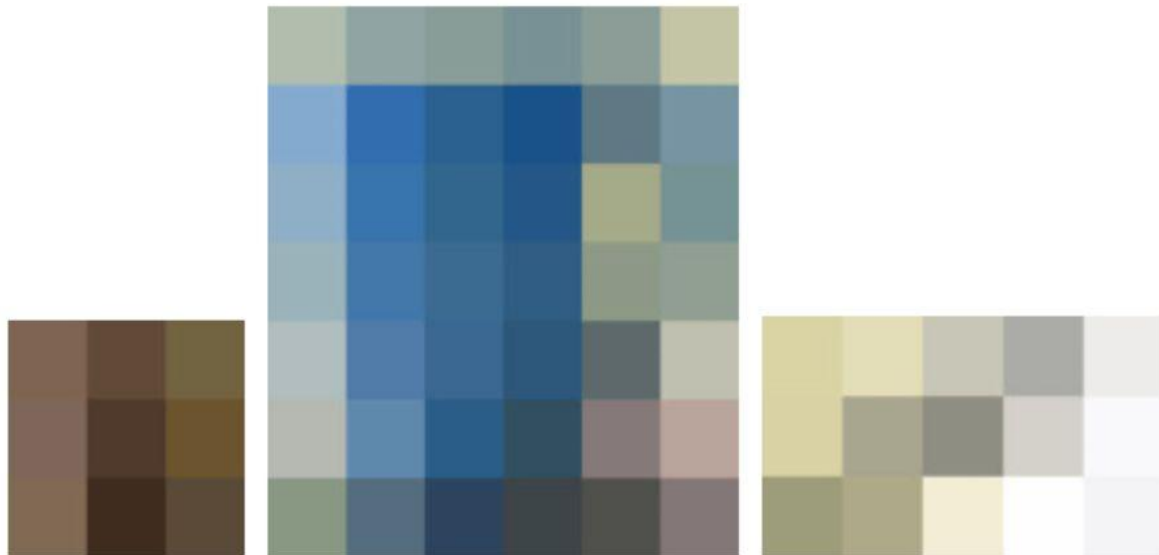
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- A **digital image** samples this function at a **finite number of locations** and represents the measurements with **finite numerical values**: a 2D array arranged in rows and columns.
- These individual elements are called **pixels** (picture elements), each with a value at a location.



A 32×32-pixel image

Context Is What Makes Pixels Meaningful

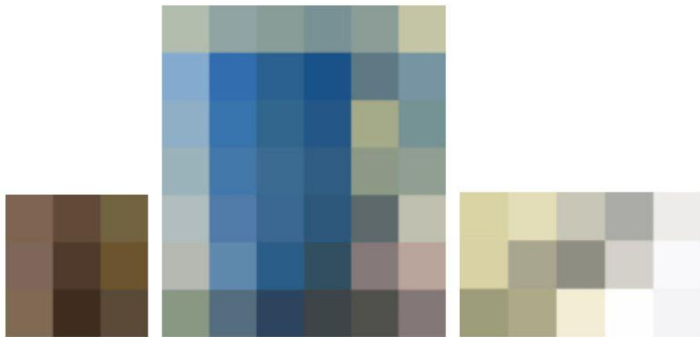
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Context Is What Makes Pixels Meaningful

? *Could you identify an object from a single, isolated pixel or a small group of pixels?*

- Taken in isolation, these crops are nearly impossible to identify. From left to right: a hand, a blue mug, and a pen.
- A pixel tells us very little by itself. **Meaning emerges from relationships between pixels.**
- So what should a computer vision model actually learn?
 - Not: Which pixel is the pen?
 - but: **Which patterns, relationships, and structures in the pixels are useful for recognizing a pen?**



Computer Vision and Society

? *When computer vision makes a mistake, what happens?*

- CV systems increasingly make decisions that affect people's lives:
 - face recognition and authentication
 - perception for autonomous driving
 - medical image analysis and clinical decision support
- As CV moves from research into the real world, errors can have real consequences:
 - a missed pedestrian → physical harm
 - a missed cancer lesion → delayed diagnosis
 - a false face match → consequences for an individual
- Performance may also differ across people or groups.
- So it is not enough to ask: **How accurate is the model?**
- We also need to ask:

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 - medical image analysis and clinical decision support
- As CV moves from research into the real world, errors can have real consequences:
 - a missed pedestrian → physical harm
 - a missed cancer lesion → delayed diagnosis
 - a false face match → consequences for an individual
- Performance may also differ across people or groups.
- So it is not enough to ask: **How accurate is the model?**
- We also need to ask: **For whom does it work, and for whom does it fail?**

Algorithmic Fairness

? *Does the model work equally well for everyone?*

- o **Algorithmic fairness** studies how algorithmic systems **perform** across **different groups of people**.
- o In practice, fairness is measured, e.g. comparing **accuracy**, **false-positive rate**, or **false-negative rate** across subgroups, rather than reporting one overall number.
- o Further reading: **MIT Visionbook — Fairness & Ethics** (visionbook.mit.edu/fairness.html).

Types of Computer Vision Tasks — Formally

Every **CV task** can be described as a **function mapping** an **image I** to some **output**.

The output's shape is what tells the tasks apart:

Task	Formal output	Meaning
Classification		
Detection		
Segmentation		

- As we move **classification** → **detection** → **segmentation**, the output gets **richer and more precise** — and both labeling and computing it get **more expensive**.

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Detection	$f(I) \rightarrow \{(b_i, c_i)\}$	a set of bounding boxes b_i , each with a class c_i
Segmentation	$f(I) \rightarrow M$	a mask M , same $H \times W$ as I , one class per pixel

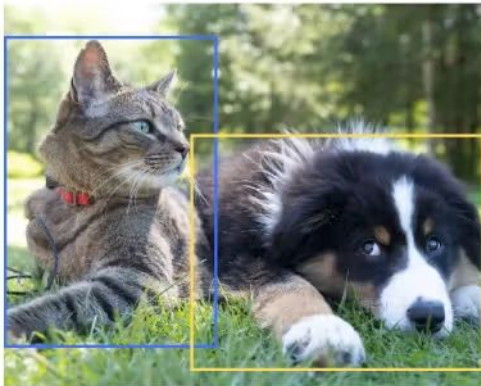
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Three Fundamental Vision Tasks

Is this a cat?



What is there in the image
and where?



Which pixels belong to
which object



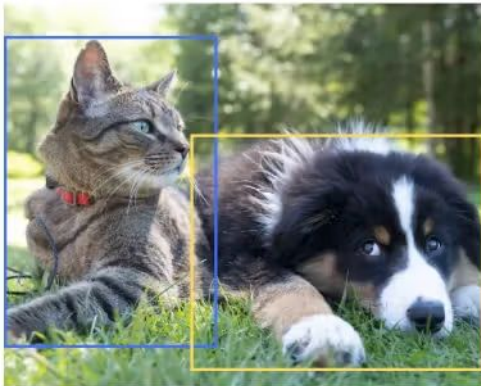
Three Fundamental Vision Tasks

Is this a cat?



Image Classification

What is there in the image
and where?



Object Detection

Which pixels belong to
which object



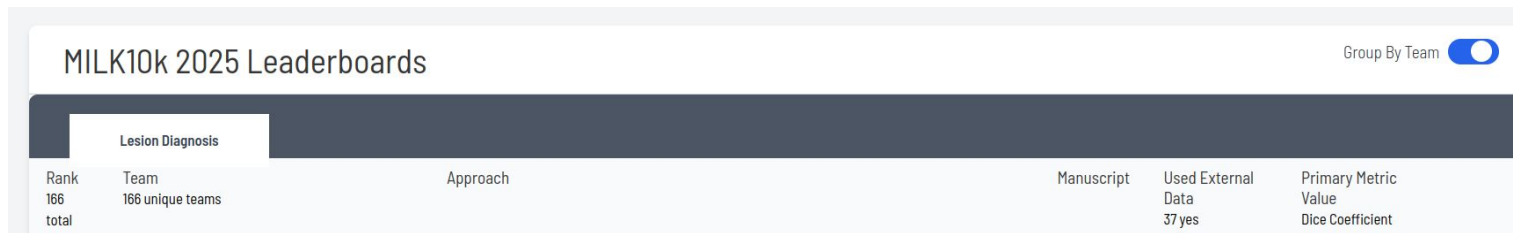
Image Segmentation

Choosing the Right Task for the Problem

- **Classification** fits screening/triage questions: “does this image show a malignant lesion?” — fast, cheap to label, one label per image.
- **Detection** fits counting/locating problems: “how many objects, and where?” — needs bounding-box labels.
- **Segmentation** fits problems needing exact boundaries: “what is the precise shape and area of this lesion?” — the most expensive to label, the most detailed output.
- Our project → MILK10k starts with **classification**.

What Is a Machine Learning “Challenge”?

- A **challenge** (or benchmark competition) is how the ML community compares methods on the **same data**, the **same task**, and the **same metric**.
- Well-known examples: **ImageNet ILSVRC** (classification), **Kaggle** competitions, and medical-imaging challenges run through the **ISIC Archive** — the same archive family MILK10k is drawn from.
- A challenge typically provides: a **labeled training set**, a **held-out test set**, an **evaluation metric**, and a **leaderboard** to rank submissions.
- Challenges drive progress by making results **comparable and reproducible**.

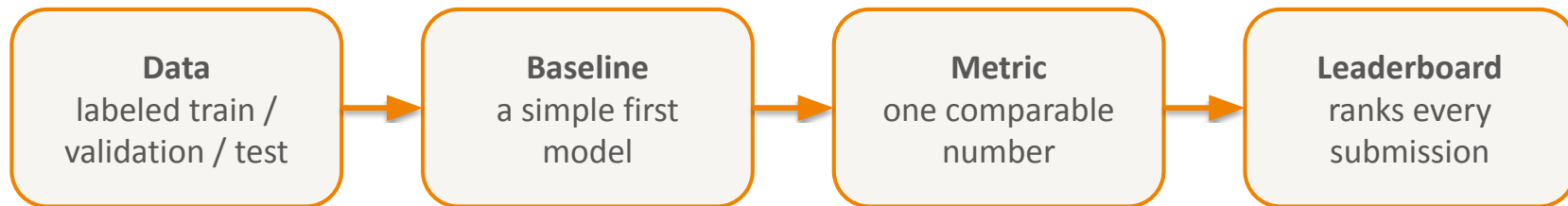


MILK10k 2025 Leaderboards Group By Team ☒

Lesion Diagnosis					
Rank	Team	Approach	Manuscript	Used External Data	Primary Metric Value
166 total	166 unique teams			37 yes	Dice Coefficient

<https://challenge.isic-archive.com/leaderboards/milk10k/>

Vision Challenge



- **Data split:** train (learn), validation (tune), test (final, unseen score).
- **Baseline:** a simple first model that sets the bar everyone tries to beat.
- **Metric:** a single number (e.g. accuracy, AUC, macro-F1) that makes results comparable.
- **Leaderboard:** ranks submissions.

Train / Validation / Test — Why Three Splits?

Train / Validation / Test — Why Three Splits?

- A model that only sees a **train** set can be checked on that same data — but a model can **memorize** training examples instead of learning general patterns. That's **overfitting**.
- The **test set** must stay unseen until the very end — it's our best estimate of how the model performs on truly new data.
- So where do we tune hyperparameters without “peeking” at the test set?

Train / Validation / Test — Why Three Splits?

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- The **test set** must stay unseen until the very end — it's our best estimate of how the model performs on truly new data.
- So where do we tune hyperparameters without “peeking” at the test set?
- The **validation set** — a second held-out slice used only during development.

Key idea

The **generalization gap** is the difference between performance on training data and on truly unseen data — the entire reason we hold data back.

Introducing MILK10k

MILK10k is the real dataset used throughout our course project — built from the ISIC Archive, the same family of skin-lesion archives behind HAM10000 and the ISIC challenges:

Quantity	Value
Images	10,480
Unique lesions	5,240 (exactly 2 images per lesion: one dermoscopic, one clinical close-up)
Diagnosis, top level	Malignant, Benign, Indeterminate

Every lesion has a stable `lesion_id` linking its two images together.

MILK10k — Real Diagnosis Categories

Per-lesion diagnosis uses an 11-class scheme (training_gt.csv).

Code	Diagnosis
BCC	Basal cell carcinoma
NV	Nevus
BKL	Benign keratosis-like lesion
SCCKA	Squamous cell carcinoma / keratoacanthoma
MEL	Melanoma
AKIEC	Actinic keratosis / intraepithelial carcinoma
DF	Dermatofibroma
INF	Inflammatory / infectious
VASC	Vascular lesion
BEN_OTH	Other benign
MAL_OTH	Other malignant

MILK10k — Demographics & Structure

Real demographic and clinical metadata, from metadata.csv:

Field	Summary
Sex	male, female
Age	mean, median, range
Anatomical site	trunk 3,850, head neck face 2,982, lower extremity 2,046, upper extremity 1,334, ...
Fitzpatrick skin tone	6-point scale, unevenly sampled — types III–IV dominate, types I & VI are rare

Course Project Overview

- Check Course Project slides.
- Two deliverables:
 - your own pipeline
 - submission to the real, official MILK10k leaderboard — **mandatory for everyone**

Part 1 Recap — Key Takeaways

- **Computer Vision** teaches machines to recover **scene meaning** from images — not just measure light.
- Images can form from many kinds of energy, not only visible light.
- Digital images are **arrays of pixels**; a single pixel rarely carries meaning without context.
- CV tasks range from **classification** to **detection** to **segmentation**.
- MILK10k is real: 10,480 images, 5,240 lesions, 11 diagnosis classes, and severe class imbalance.



TIME FOR A BREAK

20 Minutes

Stretch, grab a coffee, and be back on time.



SESSION 1

Part 2 — Hands-on Python

Loading and exploring the real MILK10k dataset

Practice Objectives

- Set up a Python environment with the libraries we'll use.
- Load MILK10k's real **metadata.csv** and **training_gt.csv** with pandas, and explore their structure.
- Load and display real **images** with PIL and matplotlib, and inspect them as NumPy arrays.
- Discover MILK10k's real class imbalance and two-images-per-lesion structure.

Homework

- **Complete the exploratory data analysis (EDA)** of the MILK10k dataset: images, metadata, class distribution, and relevant visualizations.
- Choose an **IDE/code editor** to use for the practical exercises and the project (e.g., VS Code).
- Create a **GitHub repository** for your course work and project.
- Create a **project structure** and push today's **exploratory data analysis (EDA) exercises** to your repository.
- Include a small explanation of the clinical task that we want to solve.

Downloading the Real MILK10k Dataset

? *10,480 real images, one download cell. Should that take seconds, or hours?*

One cell, run once, and the real dataset is on your Colab VM's disk:

```
# wget
# we copy the download link right-clicking on the link on the website https://api.isic-archive.com/doi/milk10k/
!wget https://isic-archive.s3.amazonaws.com/does/10.34970-648456/milk10k.zip
!unzip milk10k.zip -d milk10k
```

colab_cell

- This is the **exact, full, real** MILK10k dataset — 10,480 images, ~345 MB — not a sample. It downloads in well under a minute on Colab's connection. Run this **once per Colab session** — the VM's disk is wiped when the runtime disconnects, so you'll re-download if you reconnect later.

Loading MILK10k

? Two CSVs, two different row units. Why might that be deliberate, not an accident?

```
load_milk10k.py

from pathlib import Path
import pandas as pd

DATA_DIR = Path("milk10k")
IMG_DIR = DATA_DIR / "images"
df = pd.read_csv(DATA_DIR / "metadata.csv")           # one row per image
gt = pd.read_csv(DATA_DIR / "supplements" / "training_gt.csv")  # one row per lesion

print(df.shape, gt.shape)
print(df["lesion_id"].nunique(), "unique lesions")
```

o **df** is per-image; **gt** is per-lesion

Exploring the Metadata



explore_metadata.py

```
print(df["diagnosis_1"].value_counts())

# any missing values in key columns?
print(df[["age_approx", "sex", "anatom_site_general"]].isna().mean())

# confirm the 2-images-per-lesion structure
print((df.groupby("lesion_id").size() == 2).all())
```

- `value_counts()` is the single most useful pandas method for a first look at any label column.
- The last line should print **True** — confirming what Part 1 told you about MILK10k's structure, straight from the data.

The Real Class Imbalance, Visualized



plot_class_distribution.py

```
import matplotlib.pyplot as plt

counts = gt.drop(columns="lesion_id").sum().sort_values(ascending=False)

fig, ax = plt.subplots(figsize=(8, 4))
counts.plot(kind="bar", ax=ax, color="#F08200")
ax.set_ylabel("Number of lesions")
ax.set_title("MILK10k - class distribution (11 classes)")
plt.tight_layout()
plt.show()
```

- A bar chart makes imbalance **visible in one glance** — always plot your label distribution before training anything.
- You should see BCC towering over everything else.

Loading and Displaying a Real Image

```
row = df.iloc[0]
img_path = IMG_DIR / f'{row["isic_id"]}.jpg'

img = Image.open(img_path).convert("RGB")

plt.imshow(img)
plt.title(f'{row["diagnosis_1"]} - {row["diagnosis_2"] if pd.notna(row["diagnosis_2"]) else ""}')
plt.axis("off")
plt.show()
```

show_one_image.py

- `.convert("RGB")` guarantees 3 color channels regardless of the source file — good habit.
- **diagnosis_1** is the coarse Benign/Malignant/Indeterminate label; **diagnosis_2** is a step more specific — we'll use both across the course.

Inspecting Image Properties



inspect_array.py

```
arr = np.array(img)

print("shape:", arr.shape)      # (height, width, channels)
print("dtype:", arr.dtype)      # usually uint8
print("min / max:", arr.min(), arr.max())
```

- Loading an image with PIL and wrapping it in **np.array()** is the bridge between “a picture” and “a matrix of numbers a model can use” — exactly the F(X,Y) from Part 1.
- We study **shape**, **dtype**, and **value range** in much more depth in **Session 2**.

Building an Image-Grid Viewer

grid_viewer.py

```
def show_samples(df, img_dir, n=6, diagnosis=None, seed=0):
    subset = df if diagnosis is None else df[df["diagnosis_1"] == diagnosis]
    sample = subset.sample(n=n, random_state=seed)
    fig, axes = plt.subplots(1, n, figsize=(2.6 * n, 3))
    for ax, (_, row) in zip(axes, sample.iterrows()):
        img = Image.open(img_dir / f'{row["isic_id"]}.jpg')
        ax.imshow(img); ax.set_title(row["diagnosis_1"], fontsize=8); ax.axis("off")
    plt.tight_layout(); plt.show()

show_samples(df, IMG_DIR, n=6, diagnosis="Malignant")
```

Two Images, One Lesion

? *Split rows randomly, and could both images of one lesion land on opposite sides of train/test?*

```
lesion = df["lesion_id"].value_counts().index[0]
pair = df[df["lesion_id"] == lesion]

fig, axes = plt.subplots(1, 2, figsize=(6, 3))
for ax, (_, row) in zip(axes, pair.iterrows()):
    img = Image.open(IMG_DIR / f'{row["isic_id"]}.jpg')
    ax.imshow(img); ax.set_title(row["image_type"], fontsize=9); ax.axis("off")
plt.show()
```

two_images_one_lesion.py

- One **dermoscopic** image, one **clinical** close-up — very different statistics, same lesion.
- This is why dataset splitting must group by lesion_id, not by image row.

Practice at home

- Use today's tools to answer:
 - 1. Which `diagnosis_2` subcategory is most common within the Malignant class?
 - 2. What is the average image width and height across a 200-image sample?
 - 3. Using **`show_samples()`**, display 6 examples each of two different diagnoses. What visual differences do you notice?
 - 4. Pick one `lesion_id` and confirm it has exactly one 'dermoscopic' and one 'clinical: close-up' `image_type`.

Hint

For question 2, loop over a sample, call `Image.open(path).size`, and average the results with `np.mean()`.

Question 2



answer_q2.py

```
widths, heights = [], []  
for _, row in df.sample(200, random_state=0).iterrows():  
    with Image.open(IMG_DIR / f'{row["isic_id"]}.jpg') as im:  
        w, h = im.size  
        widths.append(w); heights.append(h)  
  
print("avg width:", np.mean(widths))  
print("avg height:", np.mean(heights))
```

- Sampling 200 images instead of all 10,480 keeps this fast — a common pattern when you just need a quick estimate.

Wrap-Up — What We Built Today

- A working Python environment with numpy, pandas, PIL, matplotlib.
- Code to load and join MILK10k's real **metadata.csv** and **training_gt.csv**.
- A reusable **show_samples()** function you'll keep using all course.
- A first, hands-on look at the real **11-way class imbalance** and the **two-images-per-lesion** structure.

Coming Up Next & References

- **Next — Session 2: Image Representation & Digital Image Fundamentals.** How an image forms and becomes numbers: acquisition, sampling, pixels, resolution, color, and normalization — theory and hands-on Python.

References & further reading

Hugging Face Computer Vision Course, Unit 1 · MIT Visionbook — Fairness & Ethics · ISIC Archive